• CHEERIO AI • PRODUCT STRATEGY MEMORANDUM

Assignment A: Scaling Customer Support for Non-Technical Users

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Executive Summary: Cheerio supports 5,000 active Indian SMB merchants generating ~17,500 inbound support queries/month. The core problem is not an FAQ deficiency; it is technical onboarding failure across Meta Cloud API compliance, phone binding, and CSV formatting. Instead of a conversational chatbot that requires users to explain technical errors, we designed **Cheerio Co-Pilot**: an embedded, state-aware widget providing 1-click deterministic remediations. This strategy reduces support cost per ticket by 62.4% (from ₹40.00 to ₹15.06), avoids 13 support hires as volume scales, generates ₹52.38L in net annual operational savings, and delivers a 50.4× net ROI on monthly AI spend by curbing merchant churn.

1.1 THE CORE SCALING BOTTLENECK FOR NON-TECHNICAL MERCHANTS

Cheerio empowers non-technical business owners (retail jewellers, direct-to-consumer apparel, clinics, and coaching institutes) to run WhatsApp marketing campaigns. However, these users struggle with strict developer platforms: Meta Business Manager, Cloud API error payloads, webhook handshakes, and strict CSV schemas. When a template is rejected or a phone number fails to bind, merchants cannot diagnose the issue. They submit repetitive, high-touch support tickets ("WhatsApp is broken", "My blast won't send"), overwhelming Cheerio's human support operations.

1.2 ONBOARDING BOTTLENECKS & FAILURE TAXONOMY (5,000 MERCHANT SCALE)

ONBOARDING BOTTLENECK	EST. VOLUME	UNDERLYING TECHNICAL CAUSE	MERCHANT EXPERIENCE & FAILURE IMPACT
Template Rejections	38% (6,650/mo)	Meta rejection `INCORRECT_CATEGORY` (promotional discount copy submitted under Utility category).	Template rejected by Meta; campaign delayed; merchant frustrated by perceived tool bug.
WABA Phone Binding	27% (4,725/mo)	Meta Error `133010` (Account not registered / phone locked to consumer WhatsApp 2FA).	Merchant stuck on phone connection; cannot broadcast; high churn risk during Day 1–3.
CSV Contact Uploads	19% (3,325/mo)	Missing +91 international dialing prefixes; illegal spaces, hyphens, and empty rows.	Broadcast campaign aborts on upload; merchant unable to fix 1,000+ Excel rows manually.
Meta Verification	11% (1,925/mo)	Two-factor auth lockouts, missing GST business proofs, or rejected display names.	Blocked by Meta Business Suite; merchant blames Cheerio for external Meta platform rules.
Webhook & Cloud Sync	5% (875/mo)	Shopify / WooCommerce sync drops, expired access tokens, catalog synchronization lags.	Abandoned cart triggers do not fire; attribution tracking breaks silently.

[Assumption: Query distribution benchmarked from Indian B2B WhatsApp BSP support tickets across onboarding and broadcast workflows; 3.5 queries/merchant/month across 5,000 businesses].

1.3 SOLUTION SPACE EXPLORATION: 8 ALTERNATIVES EVALUATED & SCORED

To avoid premature convergence on a standard chatbot, 8 distinct operational interventions were scored across Deflection Potential, Engineering Effort, User Friction, and Operational Risk:

SOLUTION ALTERNATIVE	DEFLECTION	DEV EFFORT	USER FRICTION	RISK	VERDICT / STRATEGIC FIT
1. In-App Context Co-Pilot	High (65%)	Medium (1 mo)	Near-Zero	Low	**Selected (Primary Core Feature)**
2. Proactive Field Linter	Medium (35%)	Low (2 wks)	Near-Zero	Very Low	**Hybridized into Co-Pilot frontend**
3. Guided Onboarding Checklist	Low (15%)	Low (1 wk)	Low	Low	Helpful UX, but fails to fix broken inputs
4. WhatsApp Support Bot	Medium (40%)	Medium	Medium	High	Forces copy-pasting errors between screens
5. Video & Community Library	Very Low (8%)	Low	High	Low	SMBs do not watch 10-minute documentation
6. Done-For-You (DFY) White-Glove	High (80%)	None	Zero	Severe (Cost)	Unscalable; unit economics collapse
7. Agent-Assist Internal Copilot	Zero (0%)	Medium	None	Low	Speeds agent MTTR, but defers zero tickets

• SYSTEM ARCHITECTURE & COMPLIANCE

Cheerio Co-Pilot: Technical Architecture & Operational Guardrails

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2.1 PRODUCT SPECIFICATION: THE IN-APP CONTEXT-AWARE CO-PILOT

Cheerio Co-Pilot is embedded directly into the campaign workspace (WhatsApp Templates, CSV Audiences, Phone Registration). Unlike conversational chatbots that live in a detached chat widget, Co-Pilot maintains complete real-time DOM telemetry: form field values, draft history, account verification status, and Meta Cloud API HTTP response headers. When a merchant makes an error, Co-Pilot provides two transparent, 1-click remediation paths on-screen:

Remediation Path A: Clean Utility (1-Click Auto-Fix) Cost: ₹0.1150/msg

Strips violating promotional phrases (e.g., *"HURRY! 50% OFF Diwali Sale"*) while preserving dynamic tracking variables (`{{1}}`, `{{2}}`). Rewrites the message into strict transactional compliance and updates both the template and live phone preview.

Remediation Path B: Reclassify to Marketing Cost: ₹0.8631/msg

Preserves the merchant's promotional discount copy, automatically reassigns the template category to Marketing, and alerts the merchant to the higher Meta delivery tariff (₹0.8631 vs. ₹0.1150 base rate), preventing surprise billing disputes.

2.2 MULTI-TIER RESOLUTION ENGINE & COST-OPTIMIZED ROUTING

To deliver sub-second user responsiveness and keep AI operating costs below ₹0.20 per query, Co-Pilot utilizes a 3-tier processing pipeline:

Tier 1: Client-Side Regex 0ms • Free

Deterministic client-side checks evaluate variable counts, formatting whitespace, and promotional keyword triggers. Traps 100% of missing country codes on CSV uploads without making an LLM API call.

Tier 2: Semantic Cache 25ms • ₹0.02

An Upstash Redis semantic vector cache matches inbound template copy against 2,000+ previously verified compliant templates. Returns instant cached rephrasings for standard Indian festive campaigns.

Tier 3: GPT-4o-mini 450ms • ₹0.18

Invoked only for novel, complex copy rephrasing or Hinglish intent parsing. Uses structured JSON outputs with prompt caching (50% input discount) to deliver exact replacement payloads.

2.3 VERIFIED META CLOUD API ERROR TAXONOMIES & INDIA RATE CARDS

As an official Meta Business Solution Provider (BSP), Cheerio must adhere strictly to Meta Cloud API specifications:

META ERROR / REASON	OFFICIAL META SPECIFICATION	VERIFIED RATE / REFERENCE	CO-PILOT AUTONOMOUS ACTION
INCORRECT_CATEGORY `Tag: CONTENT_MISMATCH`	Template submitted as Utility contains promotional phrases or commercial CTAs.	India Utility: **₹0.1150/msg** India Marketing: **₹0.8631/msg** *Source: developers.facebook.com*	Pre-flight linter intercepts template prior to submission; offers 1-click copy strip or category shift.
Code 133010 `ACCOUNT_NOT_REGISTERED`	Phone Number ID not registered on Cloud API; coexistence lock with mobile WhatsApp app.	Meta Registration Endpoint: `POST /{phone-id}/register` Requires 6-digit PIN	Guides merchant through 2-step verification PIN registration; triggers cloud handshake.
Code 132015 `TEMPLATE_PAUSED`	Template paused by Meta algorithms due to low recipient quality rating or spam flags.	3 to 6 hour cooling-off period before permanent disablement (Code 132016).	Informs merchant of cooling period; enforces audience segment exclusion for opt-outs.

2.4 2-FAILURE ESCALATION CIRCUIT & STRUCTURED DIAGNOSTIC TELEMETRY PACKET

If a merchant struggles or the automated fix fails **twice consecutively**, Co-Pilot trips a deterministic circuit breaker. Instead of trapping the user in an infinite loop or asking "How can I help you?", Co-Pilot compiles a structured JSON Diagnostic Packet directly to the human support queue:


```
{ "packet_uuid": "PKT-2026-CHRIO-8921", "merchant_id": "FABJEWELS_D2C", "waba_id": "99182310892", "phone_id":
"109923812903", "meta_error_code": 133010, "meta_error_subcode": "ACCOUNT_NOT_REGISTERED", "failed_attempts": 2,
"circuit_breaker_tripped": true, "client_state": { "target_phone": "+919876543210", "pin_submitted": true,
"mobile_deregistration_pending": true }, "suggested_human_action": "Initiate Meta backend provisioning reset to clear SMB
coexistence lock on phone_id 109923812903." }
```


2.5 OPERATIONAL GUARDRAILS & EXPLICIT REFUSAL BOUNDARIES

Financial & Invoicing Disputes Hard Refusal

Policy: Co-Pilot is strictly prohibited from executing financial adjustments or issuing refund guarantees. Any query regarding credit card overcharges, failed renewals, or invoice adjustments routes instantly to `finance@cheerio.in` with account billing

Meta Ban & License Boundaries Zero-Override

Policy: Cheerio cannot arbitrarily override Meta's compliance servers. When accounts are flagged for spam, Co-Pilot refuses to promise an unban. It generates the official `Meta_WABA_Appeal_Dossier.pdf` and assigns it directly to

STRATEGY, PACKAGING & GOVERNANCE

Metrics Framework, Monetization Strategy & Audit Protocols

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3.1 NORTH STAR METRIC & REALISTIC 90-DAY TRAJECTORY

Rather than chasing vanity chatbot engagement ("messages sent"), Cheerio Co-Pilot optimizes for a single customer success metric: **Zero-Touch Onboarding Completion Rate (ZOCR)**—the percentage of newly registered merchants who successfully connect their WABA, verify an audience, and schedule their first campaign without raising a human support ticket.

22.0%	33.0%	58.0%	< 5.0%
BASELINE ZOCR	DAY 90 TARGET	DAY 180 TARGET	7-DAY REOPEN RATE
[Assumed Status Quo]	+11.0% Realistic Gain	Mature Product Suite	Quality Guardrail

CO-PILOT PRODUCT INTERVENTION	TARGET ZOCR LIFT	UNDERLYING OPERATIONAL MECHANISM & VERIFICATION
Template Pre-Flight Linter	+5.5%	Eliminates 1st-attempt template rejections via real-time regex & category correction.
CSV Audience Sanitizer	+3.0%	Fixes missing +91 dial codes and spacing errors in browser before broadcast launch.
WABA Phone Registration Diagnostic	+2.5%	Guides 2-step verification PIN setup; unblocks SMB mobile coexistence traps.
Total 90-Day Achievable Expansion	**+11.0% (22% → 33%)**	**Directly unblocks 550 additional merchants during Day 1-7 onboarding.**

3.2 STRICT DEFLECTION DEFINITION & CUSTOMER QUALITY GUARDRAILS

Audit-Proof Deflection Standard: A support interaction is counted as "Deflected" *only* if the merchant completes their workflow in-app and **raises no human ticket or reopened inquiry within 7 calendar days**. Users who abandon the workflow in frustration are explicitly categorized as "Failed Deflections" to prevent false reporting.

Guardrail Metric 1: Post-Remediation CSAT — Target: ≥ 4.5 / 5.0

1-click rating ("Was this auto-fix helpful?") displayed immediately after remediation. If a specific linter rule falls below 4.0 CSAT, it is demoted to "Human Review Only" automatically.

Guardrail Metric 2: Escalation Telemetry Completeness — Target: ≥ 95%

Monitors the percentage of human-escalated tickets that contain a valid Diagnostic Telemetry Packet. Ensures support reps never ask merchants for account ID, WABA ID, or error screenshots.

4.1 MONETIZATION & BILLING STRATEGY OPTIONS (EVALUATED FROM FINANCIAL MODEL)

PACKAGING MODEL	PRICE	FULLY LOADED COST	MARGIN (%)	RECOMMENDATION & STRATEGIC RATIONALE
1. 100% Free Core Feature	₹0 / mo	₹6.26 / resolved query	Subsidized	**Recommended:** Maximizes ZOCR; fully funded by ₹1.76L/mo net retention revenue (50.4× ROI).
2. Per-Resolution Micro-Fee	₹5.00 / fix	₹0.30 (LLM+Cache)	+94.0%	Creates onboarding barrier; disincentivizes self-service; provokes SMB backlash.
3. Prepaid AI Credits Pack	₹199 / 50 fixes	₹15.00 / pack	+92.5%	High gross margin, but adds wallet friction during the critical first campaign setup.
4. Dedicated Support Add-On	₹499 / mo	₹22.50 / merchant	+95.5%	Viable optional upsell for high-volume enterprise senders needing priority SLAs.

[Fully loaded unit cost (₹6.26) = LLM token inference (₹0.20) + Observability & cache overhead (₹0.10) + Amortized human escalation share across unassisted tickets (₹5.96)].

4.2 OBSERVABILITY, SAFETY THRESHOLDS & MONTHLY AUDIT PROTOCOL

AUDIT METRIC / CHECK	ALERT THRESHOLD	SEVERITY	ASSIGNED OWNER	REMEDIATION ACTION
Cost per Resolved Query	> 2× Baseline (> ₹0.40)	High	Lead APM / ML Eng	Audit prompt token usage; revert to client regex linter.
Session Token Expansion	> 140% of 7-day avg	Medium	Backend Lead	Prune conversation history; compress error JSON payloads.

• FINANCIAL MODEL & ENGINEERING LOG

Financial Defensibility, Retrospective & Deliverables

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5.1 FINANCIAL MODEL & AVOIDED HEADCOUNT ECONOMICS (5,000 MERCHANTS)

FINANCIAL PARAMETER	PURE HUMAN MODEL	WITH CHEERIO CO-PILOT	NET DELTA / IMPACT	MODEL REFERENCE / FORMULA
Inbound Query Volume	17,500 queries / mo	17,500 queries / mo	0 (Constant)	Volume_Model!B7 (5,000 × 3.5)
Autonomous Deflection	12.0% (FAQ baseline)	65.0% (Co-Pilot target)	+53.0% Deflected	11,375 queries resolved by AI
Human Headcount Required	20 Support Agents	7 Support Agents	**-13 Avoided Hires**	Avoided hiring as volume expands
Monthly Human Payroll (₹)	₹7,00,000 / mo	₹2,45,000 / mo	-₹4,55,000 / mo	Human_Comparison!B11 vs B17 (@ ₹35k)
Monthly AI & Tooling Cost (₹)	₹0 / mo	₹18,500 / mo	+₹18,500 / mo	LLM tokens (₹3,500) + Tooling (₹15,000)
Net Monthly Support Cost (₹)	**₹7,00,000 / mo**	**₹2,63,500 / mo**	**-₹4,36,500 / mo**	**₹52.38 Lakhs / year savings**
Cost per Resolved Ticket (₹)	**₹40.00 / ticket**	**₹15.06 / ticket**	**-₹24.94 (-62.4%)**	**Human_Comparison!B12 vs B22**
CapEx Dev Build & Payback	-	₹2,50,000 build cost	**0.57 Months Payback**	~17 days payback period on build

5.2 SEPARATED CHURN ECONOMICS & NET RETENTION ROI (NO DOUBLE COUNTING)

To maintain mathematical integrity, support operational savings and retention revenue are modeled separately. With 5,000 active businesses paying ₹2,999/mo (₹1.50 Cr MRR) experiencing 5.0% baseline monthly churn (250 lost merchants):

60 Merchants
RETAINED / MONTH
1.2% Churn Reduction

₹1,76,440
NET PRESERVED MRR
Gross ₹1.79L less ₹3.5k AI

50.4× Net ROI
RETURN ON AI SPEND
Gross ROI = 51.4×

MODEL STRESS-TEST SCENARIO	CHURN REDUCTION	NET PRESERVED MRR (₹)	NET ROI MULTIPLE (ON ₹3.5K SPEND)
Conservative (Low)	0.5% absolute	₹71,475 / month	**20.4× Net ROI**
Expected (Base)	**1.2% absolute**	**₹1,76,440 / month**	**50.4× Net ROI (Gross: 51.4×)**
Aggressive (High)	2.0% absolute	₹2,96,400 / month	**84.7× Net ROI**

6.1 ENGINEERING RETROSPECTIVE & PROMPT ENGINEERING LOG

Real Prompt Iterations & Caught Failures — Audit Log

Iteration 1: Initial prompt asked for "generic chatbot ideas"—produced superficial FAQ advice. Pivoted to persona-anchored prompt on Meta BSP API failure taxonomies.

Iteration 2: Model hallucinatory output suggested overriding Meta's backend approval status. Caught and corrected: Cheerio as a BSP must lint on the frontend prior to API submission.

Candidate Plain-Voice Reflection — Candidate Voice

"I initially debated whether to charge merchants ₹5 per AI resolution to create a new profit center. However, modeling the unit economics proved that monetizing support creates friction during onboarding, causing more churn than it earns. Subsidizing Co-Pilot as a free core feature delivers a 50.4× net return through retention."

6.2 VERIFIED ASSIGNMENT DELIVERABLES & ARTIFACT ACCESS

DELIVERABLE ARTIFACT	DESCRIPTION & TECHNICAL SCOPE	ACCESS LINK / LOCATION
1. Interactive Web Prototype	4 Scenarios: Pre-Flight Linter, Circuit Breaker, Refusal, WABA Phone Binding (Code 133010) + Edge cases.	saksham-gupta-gh.github.io/cheerio-copilot/
2. Financial Model (Google Sheet)	8-tab formulaic model: Assumptions, Volume, Tokens, Headcount, Churn ROI, Billing, and Auditing.	Google Sheets Live Formula Model
3. Engineering Prompt Log	Complete transparent engineering log of prompts, token costs, hallucinations caught, and iterations.	prompt_log.md (GitHub Source)